INCH-POUND

MIL-PRF-19500/158R
2 April 2008
SUPERSEDING
MIL-PRF-19500/158P
1 July 2004

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, VOLTAGE-REFERENCE, TEMPERATURE COMPENSATED, TYPES 1N3154-1, THROUGH 1N3157-1, AND 1N3154UR-1 THROUGH 1N3157UR-1, JAN, JANTX, JANTXV, AND JANS, RADIATION HARDENED (TOTAL DOSE ONLY) JANTXVM, D, L, R, F, G, H, AND JANSM, D, L, R, F, G, H

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 8.4 volts ±5 percent, silicon, voltage-reference, temperature compensated diodes. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500. Seven levels of radiation hardened (total dose only) product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to DO-7 and DO-35) and figure 2 (DO-213AA).

1.3 Maximum ratings. Unless otherwise specified, maximum ratings (T_A = +25°C).

P_T	T_{STG} and T_J	I_{ZM} (1)	Power derating above T_A = +25°C
mW	°C	mA dc	mW/°C
500	-55 to +175	55	3.33

(1) To guarantee voltage temperature stability, it is necessary to maintain the proper I_Z = 10 mA.

1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics at T_A = +25°C.

Type	ΔV_Z (voltage-temperature stability) I_Z = 10 mA dc (1)	Z_Z I_Z = 10.0 mA dc	V_Z I_Z = 10.0 mA dc		I_R V_R = 5.5 V
			Min	Max	
	mV dc	ohms	volts	volts	µA
1N3154-1, 1N3154UR-1	130.0	15	7.98	8.82	10.0
1N3155-1, 1N3155UR-1	65.0	15	7.98	8.82	10.0
1N3156-1, 1N3156UR-1	26.0	15	7.98	8.82	10.0
1N3157-1, 1N3157UR-1	13.0	15	7.98	8.82	10.0

(1) To guarantee voltage temperature stability, it is necessary to maintain the proper I_Z = 10 mA.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA
19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and figures 1 and 2 herein.

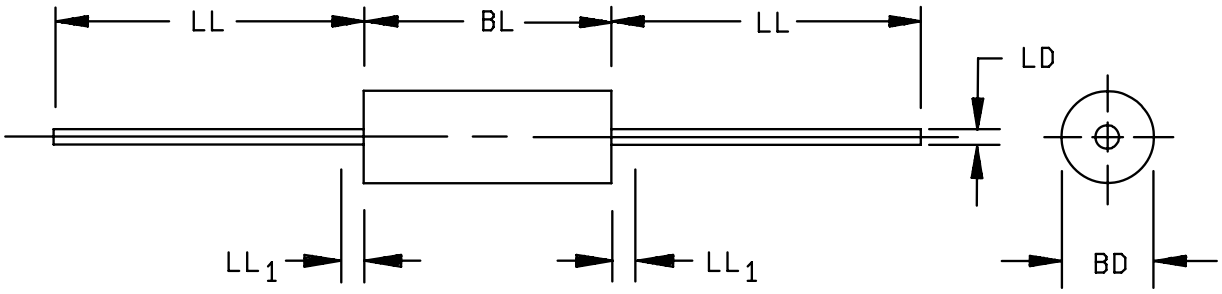
3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Diode construction. These devices shall be constructed in a manner and using material which enable the diodes to meet the applicable requirements of MIL-PRF-19500 and this document.

3.5.1 Dash-one construction. Shall be as specified in MIL-PRF-19500.

3.5.2 JANS construction. Construction shall be dash-one, category I or II metallurgical bond in accordance with MIL-PRF-19500.

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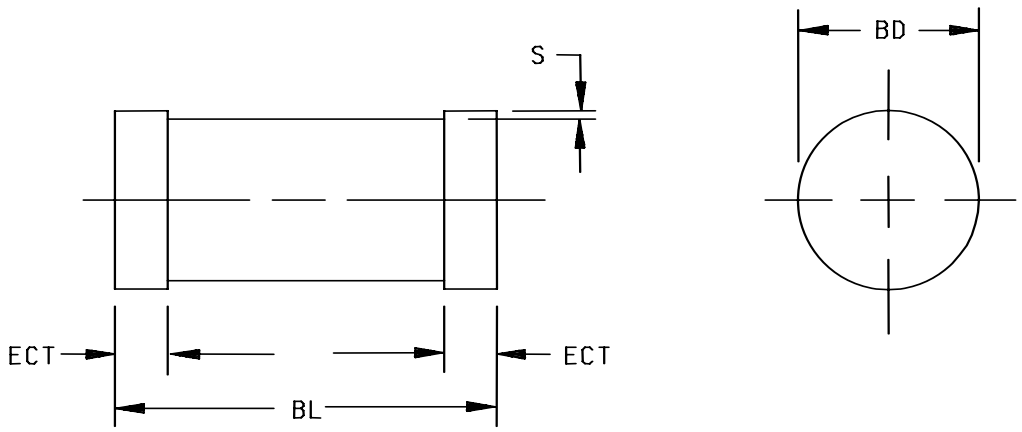


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	.060	.107	1.52	2.72	3
BL	.120	.300	3.05	7.62	3
LD	.018	.023	0.46	0.58	
LL	1.000	1.500	25.40	38.10	
LL1		.050		1.27	4

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Package contour optional within BD and length BL. Heat slugs, if any, shall be included within this cylinder but shall not be subject to minimum limit of BD.
4. Within this zone, lead diameter may vary to allow for lead finishes and irregularities, other than heat slugs.
5. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 1. Physical dimensions, 1N3154-1, 1N3155-1, 1N3156-1, 1N3157-1 (similar to DO-7 and DO-35).

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.063	.067	1.60	1.70
BL	.130	.146	3.30	3.70
ECT	.016	.022	0.41	0.56
S	.001 Min		0.03 Min	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 2. Physical dimensions, 1N3154UR-1, 1N3155UR-1, 1N3156UR-1, and 1N3157UR-1 (DO-213AA).

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3.6 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6.1 Marking of "UR" version devices. For "UR" version devices only, all marking (except polarity) may be omitted from the body, but shall be retained on the initial container.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.1.1 Sampling inspection. Sampling inspection shall be in accordance with MIL-PRF-19500 and as specified herein, except that lot accumulation period shall be 3 months in lieu of 6 weeks.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500 and 4.4.5 herein.

4.2.2 Radiation hardened devices. See MIL-PRF-19500 and 4.4.4 herein.

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4.3 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with appendix E, table E-IV of MIL-PRF-19500, and as specified herein. Specified electrical measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	JANS level	JANTXV and JANTX level
1a 1b	Required Required	Not required Required (JANTXV only)
2	Not required	Not required
3a 3b 3c	Required Not applicable Not applicable	Required Not applicable Not applicable
4	Not applicable	Not applicable
5	Not applicable	Not applicable
6	Not applicable	Not applicable
7a 7b	Not applicable Optional	Not applicable Optional
8	Required	Not required
9	Required	Not applicable
10	Required	Not applicable
11	Required V_Z, Z_Z	Required V_Z, Z_Z
12	Required see 4.3.1	Required see 4.3.1
13	Required Subgroups 2 and 3 of table I herein; $\Delta Z_Z \leq \pm 15$ percent of initial reading. T_A = +25°C ±2°C, $\Delta V_Z \leq \pm 0.004$ V dc from initial value for 1N3154-1, 1N3154UR-1, 1N3155-1, 1N3155UR-1. $\Delta V_Z \leq \pm 0.003$ V dc from initial value for 1N3156-1, 1N3156UR-1, 1N3157-1, 1N3157UR-1.	Required Subgroups 2 of table I herein; $\Delta Z_Z \leq \pm 15$ percent of initial reading. T_A = +25C ±2°C, $\Delta V_Z \leq \pm 0.004$ V dc from initial value for 1N3154-1, 1N3154UR-1, 1N3155-1, 1N3155UR-1. $\Delta V_Z \leq \pm 0.003$ V dc from initial value for 1N3156-1, 1N3156UR-1, 1N3157-1, 1N3157UR-1.
14a 14b	Not applicable Required (1) (2)	Not applicable Required (1) (2)

1/ See MIL-PRF-19500.
2/ For clear glass diodes, the hermetic seal (gross leak) may be performed at any time after temperature cycling.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: I_Z = 10 mA dc min, ± .75 mA dc, T_A = +150°C, +5°C, -0°C.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table E-V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIa (JANS) and table E-VIb (JAN, JANTX, JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable steps of table I, subgroup 2 herein.

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4.4.2.1 Group B inspection, table E-VIa (JANS) of MIL-PRF-19500. For purposes of JANS inspection, a single device type shall be defined as devices from a single wafer lot (for each die type used in the construction). The conformance inspection sample shall be selected from the part category with the lowest ΔV_Z rating in the inspection lot.

Subgroup	Method	Conditions
B1	2066	As specified.
B2	2026	As specified.
B2	1022	As specified.
B3	1056	Test condition A, 25 cycles.
B3	4066	Not applicable.
B3	1071	Test condition E.
B3	2075	As specified.
B4	1037	I_Z = 27.5 mA dc at T_A = room ambient; t_{on} = toff = 30 seconds minimum for 4,000 cycles. Forced air cooling allowed during off cycle.
B5	1027	I_{ZM} = 55 mA dc for 96 hours. T_A = +75°C, adjust T_A to achieve T_J = +200°C minimum.
B6		Not applicable.

4.4.2.2 Group B inspection, table E-VIb (JAN, JANTX, JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B1	2026	As specified.
B1	1022	As specified.
B2	1056	Test condition A, 25 cycles.
B2	4066	Not applicable.
B2	1071	Test condition E.
B3	1027	See 4.3.1.
B4	2075	As specified.
B5		Not applicable.
B6	1032	As specified.

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4.4.3. Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable steps of table I, subgroup 2 herein.

4.4.3.1 Group C inspection, appendix E, table E-VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C1	2066	As specified.
C2	1056	Test condition A, 25 cycles.
C2	2036	Lead tension: Test condition A; 4 pounds weight, t = 15 ±3 seconds. Lead fatigue: Test condition E (lead tension and fatigue tests are not applicable to surface mount "UR" version devices).
C2	1071	Condition E.
C2	1021	Omit initial conditioning.
C3		Not applicable.
C4	1041	As specified.
C5		Not applicable.
C6	1026	I_Z = 10.0 mA dc, T_A = +100°C (see 4.5.2).
C7		Not applicable.

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with appendix E, table E-VIII of MIL-PRF-19500 and table II herein. Submitted lots for group D sample inspection must be constructed using one homogeneous wafer lot for the zener and one wafer lot for the compensating die (die), as also described in the submitted DSCC Design and Construction form 36D (see table II herein).

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with appendix E, table E-IX of MIL-PRF-19500 and table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Voltage-temperature stability. The breakdown voltage of each diode type shall be measured and recorded at each of the specified temperatures. The lowest measured voltage shall be subtracted from the highest measured voltage for each diode. The difference value obtained shall not exceed the specified ΔV_Z per diode type.

4.5.2 Reference voltage time stability. The breakdown voltage shall be measured prior to life testing, at 340 hours, and at the conclusion of the life test. The 340-hour reading shall be compared with the 0-hour reading and the 1,000-hour reading compared with the 340-hour reading. The change in breakdown voltage shall not exceed the limits specified (see table IV herein).

4.5.3 Reference voltage. The test current shall be applied until thermal equilibrium is attained (15 seconds minimum) prior to reading the reference voltage. For this test, the diode shall be suspended by its leads with mounting clips whose inside edge is located between .375 inch (9.53 mm) and .500 inch (12.70 mm) from the body and the mounting clips shall be maintained at the specified temperature. This measurement may be performed after a shorter time following application of the test current than that which provides thermal equilibrium if correlation to stabilized readings can be established to the satisfaction of the Government.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Reference voltage	4022	I_Z = 10.0 ±0.01 mA dc (see 4.5.3).	V_Z	7.98	8.82	V dc
Small signal breakdown impedance	4051	I_Z =10.0 +0.1 mA dc, Isig = 1.0 mA ac.	Z_Z		15.0	ohms
Subgroup 3						
Voltage-temperature stability		I_Z = 10.0 ±0.01 mA dc, T_A = -55°C, 0°C. +25°C, +75°C, +100°C ±2°C. (see 4.5.1 and 4.5.3)	ΔV_Z			
1N3154-1, 1N3154UR-1					130.0	mV dc
1N3155-1, 1N3155UR-1					65.0	mV dc
1N3156-1, 1N3156UR-1					26.0	mV dc
1N3157-1, 1N3157UR-1					13.0	mV dc
Subgroups 4, 5, and 6						
Not applicable						
Subgroup 7						
Reverse current leakage	4016	DC method; V_R = 5.5 V dc.	I_R		10.0	µA

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Group D inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Pre-irradiation limits		Post-irradiation limits		ΔV_{Z1} pre-post irradiation change		Unit
	Method	Conditions		M, D, R, L, F, G and H		M, D, R, L, F, G and H				
				Min	Max	Min	Max	Min	Max	
Subgroup 1										
Not applicable										
Subgroup 2		T_C = +25°C								
Steady-state total dose irradiation	1019	I_Z = 10.0 ± 0.01 mA dc, condition A								
Reference voltage	4022	I_Z = 10.0 ± 0.01 mA dc (see 4.5.3)	V_Z	7.98	8.82	7.98	8.82			V dc
Small-signal breakdown impedance	4051	I_Z = 10.0 ± 0.01 mA dc I_{sig} = 1.0 ac	Z_Z		15		15			ohms
Revere current leakage	4016	DC method; V_R = 5.5 V dc	I_R		10		10			µA
Voltage stability	4022	I_Z = 10.0 ± 0.01 mA dc T_A = +25°C ± 2°C (see 4.5.1)	ΔV_Z							
1N3154-1, 1N3154UR-1									± 4.0	mV
1N3155-1, 1N3155UR-1									± 3.5	mV
1N3156-1, 1N3156UR-1									± 3.0	mV
1N3157-1, 1N3157UR-1									± 2.5	mV

1/ For sampling plan, see MIL-PRF-19500.
2/ Group D qualification may be performed any time prior to lot formation.

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TABLE III. Group E inspection qualification and requalification (all product assurance levels).

Inspection	MIL-STD-750		Qualification conformance inspection (sampling plan)
	Method	Conditions	
Subgroup 1			22 devices, c = 0
Temperature cycling	1051	500 cycles.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			22 devices, c = 0
Steady-state operation life	1038	1,000 cycles. See 4.3.1.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 3			3 devices, c = 0
Decap analysis	2101	Cross section and scribe and break. Separate samples shall be used for each test.	
Subgroups 4, 5, 6, and 7			
Not applicable			
Subgroup 8			45 devices
Resistance to glass cracking	1057	Step stress to destruction by increasing cycles or up to a maximum of 25 cycles.	

TABLE IV. Reference voltage time stability.

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Reference-voltage time stability		T_A = +100°C ±2°C (see 4.5.2 and 4.5.3), I_Z = 10 ± 0.01 mA dc	ΔV_Z			
1N3154-1, 1N3154UR-1		(0 to 340 hours)			7	mV dc
1N3155-1, 1N3155UR-1					7	mV dc
1N3156-1, 1N3156UR-1					6	mV dc
1N3157-1, 1N3157UR-1					5	mV dc
1N3154-1, 1N3154UR-1		(340 to 1,000 hours)			4	mV dc
1N3155-1, 1N3155UR-1					4	mV dc
1N3156-1, 1N3156UR-1					3	mV dc
1N3157-1, 1N3157UR-1					3	mV dc

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil .

6.4 Substitution of radiation hardened devices. See MIL-PRF-19500.

* 6.5 Substitutability of dash one parts. Non-dash-one devices have been deleted from this specification. Dash-one devices are a direct substitute for non dash-one devices and are preferred. The following table shows the direct substitutability.

Superseded PIN	Superseding PIN
1N3154	1N3154-1
1N3155	1N3155-1
1N3156	1N3156-1
1N3157	1N3157-1

6.6 Substitution of devices. Device types within this series with higher type numbers (lower ΔV_Z voltage-temperature stability) are a direct one-way substitution for lower type numbers (higher ΔV_Z voltage-temperature stability).

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6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
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Preparing activity:
DLA - CC

(Project 5961-2008-011)

Review activities:
Army - AR, MI, SM
Navy - AS, MC
Air Force - 19, 99

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